



An adaptive intrusion detection framework for mobile ad-hoc networks using feature extraction and NS3 simulation

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Abstract

Mobile Ad-Hoc Networks (MANETs) are inherently dynamic and infrastructure-less, making them susceptible to a range of security threats such as black hole and wormhole attacks. This research presents a novel and adaptive Intrusion Detection System (IDS) tailored for MANETs, combining Binary Relevance Strategy (BRS) and machine learning for efficient and accurate intrusion detection. The proposed framework incorporates feature extraction from NS-3 simulated network traffic, focusing on 21 crucial features to distinguish normal from malicious behaviors. It introduces a Fuzzy Logic System (FLS) to evaluate performance reliability, enabling adaptive sensitivity to network dynamics and resource constraints like energy and memory limitations. Unlike traditional IDS solutions, this system emphasizes real-time detection with reduced false positives and minimal computational overhead. The novelty of this work lies in its integration of BRS for multi-label classification, feature-optimized lightweight IDS design, and FLS-based adaptability, resulting in a scalable and energy-aware security solution. Comprehensive simulations using NS-3 validate the framework's superiority in maintaining high detection accuracy and operational efficiency, making it suitable for deployment in critical MANET applications.

Keywords Mobile Ad-hoc networks · Intrusion detection · Binary relevance strategy · Fuzzy logic system

1 Introduction

Mobile Ad-Hoc Networks (MANETs) have emerged as a crucial innovation in wireless communication, offering self-organized, infrastructure-free, and dynamic network solutions particularly suited for environments with limited or no existing infrastructure, such as disaster recovery zones, military operations, and emergency response scenarios [1] (Fig. 1). MANETs allow direct, flexible communication by forming a temporary network where devices can act as both hosts and routers, transmitting data packets across multiple hops [2]. This structure eliminates the need for a centralized controller, thus granting agility and operational flexibility. However, this very openness and lack of permanent architecture make MANETs more susceptible to security threats, including eavesdropping, data interception, packet dropping,

and various forms of attacks such as black hole, wormhole, and jellyfish attacks [3].

As MANET nodes frequently shift positions, their topology changes dynamically, leading to higher vulnerability levels and increased challenges for security maintenance [4]. Intrusion Detection Systems (IDS) in MANETs aim to address these challenges by identifying malicious activities and mitigating threats through real-time monitoring of network traffic [5]. These IDSs play a fundamental role in protecting the network, often employing cryptographic methods to encrypt data packets or relying on authentication protocols to secure communications. Cryptographic techniques, though effective, tend to increase computational complexity and resource consumption, thus prompting a shift toward more efficient, lightweight IDS methods that leverage machine learning (ML) techniques for quicker and more adaptive threat detection [6].

Among the multiple IDS strategies, machine learning and deep learning approaches have shown promising potential in efficiently detecting intrusions with minimal computational overhead [7]. Supervised feature selection, for instance, is one technique that allows IDSs to focus on critical features of network data, thus optimizing the detection process without straining the network's memory or energy resources [8]. A

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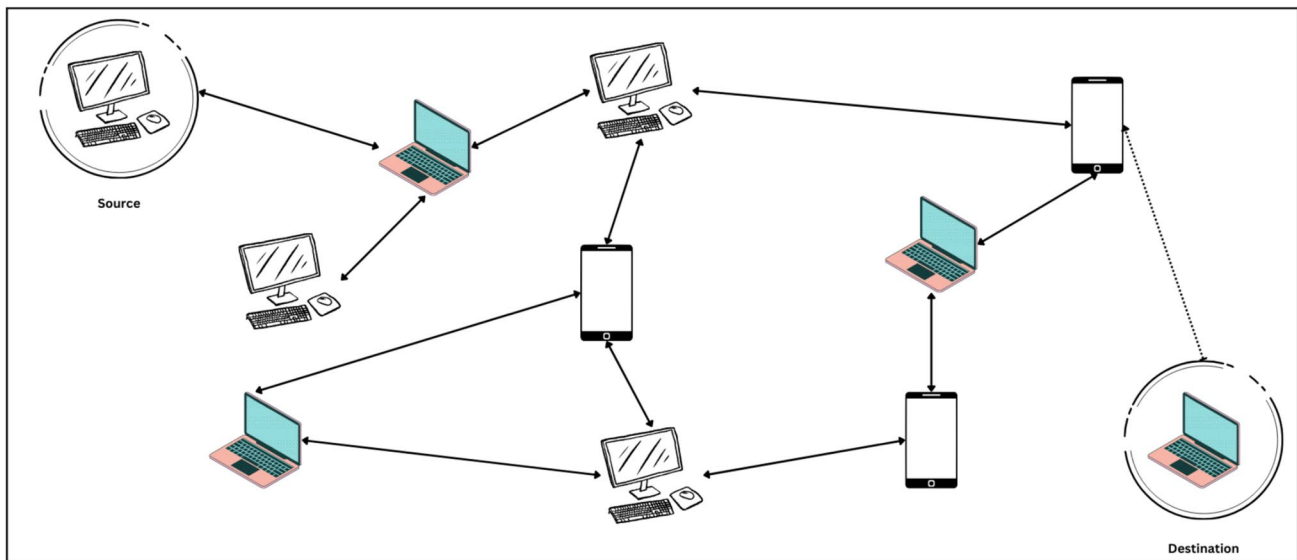


Fig. 1 MANET Architecture

more advanced, "heavyweight" IDS approach incorporates large feature sets and uses robust ML algorithms such as Multilayer Perceptron (MLP) and neural networks to perform comprehensive analyses on extensive data streams, enhancing the detection accuracy of complex attack patterns [9]. Nonetheless, such approaches may encounter scalability issues, as increased data processing demands translate into significant power and memory consumption, which are often limited in MANET environments.

In addition to intrusion detection, performance reliability evaluation is essential for understanding how IDS mechanisms can consistently function under varying network conditions. Reliable IDS methods should not only detect intrusions accurately but also operate efficiently within the constraints of limited battery life and computational power in MANET nodes [10]. The concept of a reliability evaluation model, therefore, incorporates parameters like energy consumption and computational load to gauge IDS performance under real-world conditions. One promising approach is using Fuzzy Logic Systems (FLS) to adaptively manage these constraints and achieve a balanced performance in IDS operations [2].

Routing protocols also influence IDS effectiveness within MANETs. Common routing protocols, such as Ad hoc On-Demand Distance Vector (AODV), Optimized Link State Routing (OLSR), and Destination-Sequenced Distance-Vector (DSDV), play a role in determining network behavior and potential vulnerabilities [11, 12]. Research comparing these protocols suggests that each has unique strengths and weaknesses in handling dynamic network topologies [13], with AODV and OLSR typically performing better in high-mobility scenarios due to their adaptable path discovery mechanisms [14]. However, the choice of protocol affects

how effectively IDSs can monitor and respond to attacks, as different protocols handle packet forwarding and route maintenance in varying ways, influencing the detection accuracy of IDS [15].

Given the array of IDS mechanisms and routing protocols available, recent studies have focused on simulating network performance and attack scenarios using tools like NS3 [16]. These simulations allow researchers to analyze MANET protocols and IDS configurations in controlled environments, generating data on packet loss, delay, and overall network performance under attack conditions [17]. Simulation-based evaluations provide insights into the resilience of IDSs and routing protocols, especially against common threats like black hole and wormhole attacks [9]. Notably, enhanced security techniques, such as baiting and timer mechanisms, have been developed to counter specific attacks, thus improving the reliability of standard protocols like AODV in hostile environments [14].

Despite these advancements, existing IDS frameworks often overlook the need for a comprehensive performance reliability model that accounts for the real-world demands on network nodes [15]. While many studies prioritize detection accuracy or reduction in false alarms, they seldom address how IDS performance scales in relation to resource consumption. This gap underscores the need for IDSs that are not only effective but also sustainable in terms of energy and memory utilization across diverse MANET scenarios. Such a focus would allow MANETs to support sustained, secure communication even under resource constraints, further enhancing their suitability for critical applications in military and emergency response contexts [9].

This paper addresses these challenges by proposing an enhanced IDS framework coupled with a performance

reliability evaluation model, designed specifically for MANET environments. By analyzing a carefully selected feature set, the proposed IDS can differentiate between malicious and legitimate node behavior with high accuracy. Additionally, the use of a Fuzzy Logic System in our evaluation model allows for adaptive management of resource constraints, providing a balanced approach to intrusion detection that maintains both efficacy and efficiency. Simulation results from NS3 show that the proposed IDS not only achieves superior intrusion detection but also sustains reliable performance in energy-constrained environments, thus establishing a more resilient defense mechanism for MANETs under diverse operational scenarios.

In exploring the security landscape of MANETs, understanding the specific types of attacks and their impact on network performance is crucial. MANETs face a variety of security threats beyond black hole and wormhole attacks, including gray hole, Sybil, and denial of service (DoS) attacks, which target the network's integrity, availability, and confidentiality [4, 18]. Each type of attack exploits the network's decentralized nature, making it essential for IDS mechanisms to dynamically adapt to these threats and provide timely, accurate detection. Black hole attacks, for instance, involve malicious nodes misleading other nodes to reroute data through them, which then results in packet drops and information loss. In contrast, wormhole attacks create unauthorized tunnels in the network, enabling attackers to intercept and manipulate data packets, often leading to substantial data breaches and delays in communication [18].

Intrusion detection systems for MANETs must therefore incorporate mechanisms that address the unique challenges of such attacks while remaining lightweight and efficient. To achieve this, IDS design has increasingly turned to machine learning and artificial intelligence to refine detection and classification capabilities. For instance, feature selection techniques allow IDSs to narrow down to the

most significant attributes of network traffic, thus reducing computational load and focusing on relevant data to make faster, more accurate decisions [8]. In particular, supervised learning algorithms like Bayesian classifiers, support vector machines (SVMs), and neural networks have shown promise in achieving high detection accuracy for complex attack patterns by classifying patterns as either malicious or benign based on previous data [7].

The integration of machine learning into IDS also brings challenges, particularly in balancing detection accuracy with computational efficiency. While larger feature sets may improve detection rates, they simultaneously increase energy and processing demands, a trade-off that can undermine network performance in energy-constrained MANET environments [10] (Fig. 2). Thus, optimization techniques are often applied alongside machine learning to manage the trade-offs. For example, ant colony optimization and genetic algorithms are explored for routing security, offering ways to dynamically adjust routes to avoid compromised nodes while conserving resources [19]. These approaches align with the self-organizing principles of MANETs, ensuring IDS can operate effectively without excessive resource consumption.

Despite the effectiveness of machine learning-based IDSs, they are not without limitations. Complex models, such as deep learning architectures, require extensive training data and computational power, which may not be feasible in real-time or resource-limited MANET environments. To mitigate this, recent studies have proposed hybrid approaches that combine lightweight detection rules with periodic deep analysis, allowing the IDS to perform quick initial checks with minimal resource use and then deeper analysis as needed [2]. Furthermore, fuzzy logic systems (FLS) have been incorporated into IDS to enable adaptive, threshold-based intrusion detection that can adjust sensitivity based on network conditions and resource availability [2].

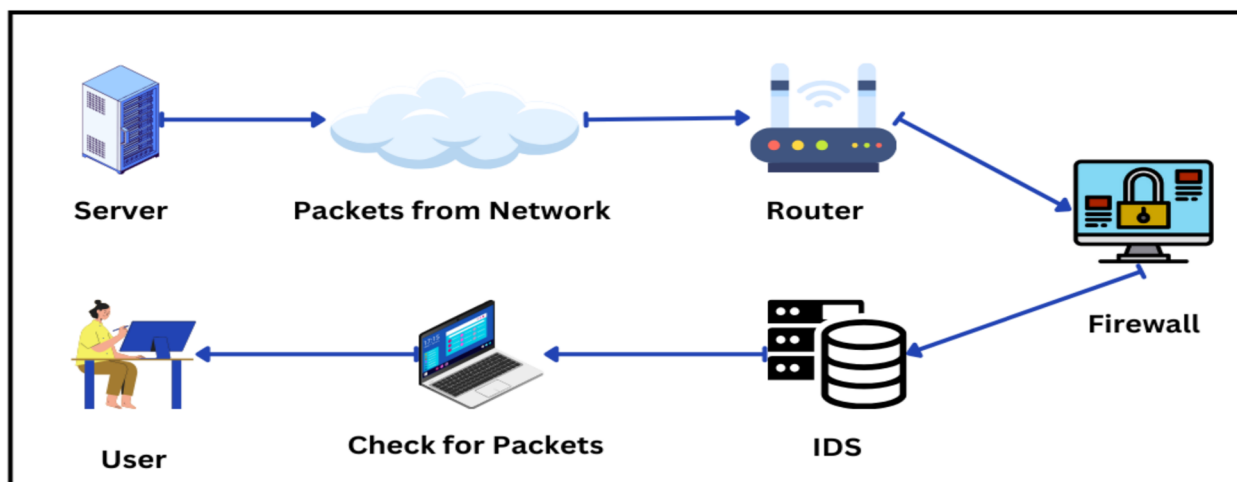


Fig. 2 IDS Architecture

Performance evaluations in simulated environments like NS3 provide valuable insights into the practical effectiveness of these IDS mechanisms. NS3 enables comprehensive testing under controlled conditions, simulating various routing protocols such as AODV, OLSR, and DSDV to assess how IDS strategies perform across diverse network topologies and traffic patterns [11, 17]. Through these simulations, researchers can measure key performance indicators such as packet delivery ratio, end-to-end delay, and packet loss under different attack scenarios, revealing the strengths and weaknesses of each IDS configuration. For instance, studies have shown that AODV, while highly adaptable, may be more susceptible to black hole attacks compared to OLSR, which uses a more stable routing mechanism but may not perform as well under high-mobility conditions [14].

In addition to the routing protocol, the reliability of IDS is often influenced by the specific security mechanisms in place. Cryptography-based approaches, though highly secure, often demand significant computational resources, making them less ideal for MANETs with limited node capabilities [6]. Lightweight IDS mechanisms, therefore, offer a more practical alternative by using simplified rules, such as monitoring packet forwarding rates, to quickly flag potential intrusions with minimal computational overhead [9]. This approach, however, may yield lower accuracy compared to more complex models, highlighting the ongoing need for IDS solutions that balance robustness with operational efficiency.

This study further extends the field by proposing a multi-layer neural network-based IDS that effectively classifies network behavior with an optimized feature set, reducing memory usage while maintaining high detection rates. In doing so, it builds on previous work by incorporating a performance reliability model using fuzzy logic, which evaluates IDS performance in terms of energy and memory consumption alongside traditional metrics such as accuracy and false-positive rate [1]. By accounting for these operational constraints, the proposed model offers a more comprehensive assessment of IDS capabilities, aligning with real-world requirements for sustainable and resilient network security.

As MANETs continue to evolve, particularly in mission-critical applications like military communications and disaster response, the importance of reliable, efficient intrusion detection cannot be overstated. Effective IDS not only safeguard sensitive information but also ensure the stability and functionality of the network itself, preventing disruptions that could impact life-saving operations [9]. With the development of more adaptable and resource-conscious IDS, future research can focus on expanding these solutions to accommodate the diverse and evolving landscape of MANET applications, promoting secure, resilient networks capable of withstanding a wide range of security threats.

One of the critical aspects of intrusion detection in MANETs is the need for adaptive solutions that can

operate effectively under the network's inherently dynamic and resource-constrained environment. MANETs' decentralized nature, combined with their high node mobility, presents unique challenges for ensuring timely and accurate intrusion detection. Traditional IDS approaches, often designed for static or infrastructure-based networks, lack the flexibility to respond to the fluctuating topologies and bandwidth constraints of MANETs, thereby necessitating tailored IDS solutions that prioritize adaptability and real-time responsiveness [5, 8].

To address these challenges, researchers have increasingly turned toward hybrid IDS approaches that combine rule-based detection with statistical or machine learning-based methods. These hybrid systems leverage the strengths of each approach, using rule-based methods for quick detection of known attack patterns while applying machine learning models for more complex, evolving threats. For instance, simple rule-based detection mechanisms can identify anomalies in packet forwarding rates or sudden changes in node behavior, which often serve as preliminary indicators of malicious activity [3]. At the same time, supervised learning techniques, such as neural networks or decision trees, can process more intricate data patterns to distinguish between benign anomalies and sophisticated attacks, such as those caused by Sybil nodes or colluding attackers [6].

The increased reliance on machine learning within IDS frameworks brings with it considerations regarding feature selection and model interpretability. As MANETs may lack the computational power to support deep learning models with extensive input features, the careful selection of a minimal yet effective feature set is crucial. Methods like rough set theory and Bayesian approaches have been explored to identify the most relevant features, which allow the IDS to process data more efficiently without compromising detection accuracy [7]. Additionally, interpretability remains essential for validating model decisions, especially in critical applications like military or emergency networks, where transparent decision-making processes are required to maintain user trust and operational reliability [20].

In recent years, unsupervised learning methods, such as clustering and anomaly detection algorithms, have also emerged as promising additions to the IDS toolkit. Unlike supervised methods, which require labeled datasets, unsupervised approaches detect patterns or deviations within unlabeled data, making them particularly well-suited for identifying previously unknown attack types or emerging threat patterns. For example, clustering techniques like K-means or DBSCAN can group similar node behaviors, flagging those that deviate significantly as potential intrusions [3]. These methods are valuable for MANET environments, where new and evolving threats may arise rapidly, outpacing traditional IDS updates.

Energy efficiency is another primary concern in MANET IDS design. Since MANET nodes are typically

battery-powered, a detection system's energy requirements can significantly impact network longevity and performance. Many IDS solutions now integrate energy-aware algorithms that reduce computational overhead by performing detection in stages or focusing only on high-risk nodes rather than the entire network [15]. For instance, an initial lightweight detection phase may scan for broad anomalies with minimal resource consumption, triggering a more in-depth analysis only when certain thresholds are exceeded. This tiered approach conserves energy while still maintaining a high level of security, making it an ideal solution for energy-limited environments [18].

Simulation platforms such as NS3 play an instrumental role in testing IDS effectiveness in MANETs by replicating various real-world scenarios and assessing system performance under diverse conditions [15]. NS3 enables researchers to evaluate IDS under simulated attacks, routing protocols, and network configurations, providing insights into the IDS's robustness and reliability. Studies using NS3 have demonstrated the impacts of protocol-specific vulnerabilities, such as those found in AODV or OLSR protocols, helping to fine-tune IDS parameters and detect vulnerabilities specific to these routing mechanisms [11, 21]. Such simulation-driven evaluations provide a comprehensive understanding of IDS behavior in real-time, allowing researchers to refine detection strategies based on measurable outcomes.

To further enhance IDS robustness, many researchers are exploring secure routing protocols that inherently prevent certain types of attacks or malicious behaviors. Protocols like AODV have been modified with security enhancements to resist black hole and wormhole attacks, using techniques such as homomorphic encryption and baiting tactics to secure data transmission and detect suspicious nodes [4, 9]. While these security-modified protocols do not replace IDS, they complement detection efforts by reducing the network's exposure to attacks, creating a layered security approach. In scenarios where secure routing protocols and IDS are deployed together, the overall resilience of MANETs against intrusions and network disruptions is significantly strengthened [2].

In this paper, we contribute to the advancement of intrusion detection systems (IDS) in Mobile Ad-Hoc Networks (MANETs) by integrating fuzzy logic to enhance adaptability and accuracy. Fuzzy Logic Systems (FLS) introduce flexible detection parameters that adjust dynamically to changes in network conditions and attack intensity. This probabilistic decision-making capability is particularly effective in MANETs, where traditional rule-based detection methods often fall short due to high variability in network behavior. By scaling detection thresholds based on factors such as node density, signal strength, and traffic volume, FLS-enhanced IDS frameworks can improve detection accuracy while reducing false positives under dynamic network scenarios [2].

Another promising avenue explored in this work is the application of ensemble learning techniques in IDS

frameworks for MANETs. Ensemble methods such as bagging, boosting, and stacking combine predictions from multiple machine learning models to increase overall robustness and accuracy. These techniques are particularly effective at identifying complex and multi-stage attacks that individual models might miss. Moreover, the inherent redundancy in ensemble learning provides fault tolerance, which is beneficial in MANET environments where node failures or disconnections are common. The proposed work aims to bridge existing gaps by using a binary relevance strategy for IDS, which operates efficiently within MANET constraints, addressing real-time adaptability needs.

The major contributions of this paper are as follows:

- In our IDS framework that integrates Fuzzy Logic Systems to dynamically adapt detection sensitivity based on real-time network metrics. This proposed model based on Machine learning approach to improve intrusion detection accuracy and resilience in highly dynamic MANET environments.
- A comprehensive simulation-based evaluation using NS-3, demonstrating the effectiveness of the proposed method in reducing false positives and increasing detection accuracy.

1.1 Motivations and challenges

The key motivation for developing adaptive IDS frameworks in MANETs is the pressing need to ensure secure communication under resource-constrained, decentralized, and highly dynamic environments. Mobile Ad-Hoc Networks (MANETs) are typically deployed in mission-critical applications such as military operations, disaster recovery, and emergency response, where network security and reliability are paramount. In such environments, traditional intrusion detection mechanisms, which often rely on fixed infrastructure and static configurations, prove ineffective. The self-configuring nature of MANETs, coupled with the lack of centralized infrastructure, introduces vulnerabilities that malicious actors can exploit.

Additionally, the unpredictability of node mobility and wireless communication exposes the network to frequent disconnections and route changes, making it essential to develop Intrusion Detection Systems (IDS) that can operate efficiently and autonomously. Another driving factor is the constrained nature of MANET nodes, which operate with limited battery power, bandwidth, and processing capabilities—necessitating lightweight and intelligent IDS mechanisms. The growing sophistication and frequency of cyber-attacks further highlight the need for proactive, learning-based defense strategies that can adapt to new and evolving threats with minimal manual intervention.

The following challenges drive the need for intelligent IDS solutions:

- **High Mobility and Dynamic Topologies:** Continuous node movement disrupts routing paths and degrades performance, complicating consistent intrusion monitoring.
- **Lack of Centralized Control:** The absence of centralized monitoring systems demands distributed and autonomous security mechanisms.
- **Exposure to Multiple Attack Types:** MANETs are susceptible to diverse attacks like Blackhole, Wormhole, Grayhole, Sybil, and DoS.
- **Real-Time Responsiveness:** Intrusions must be detected and mitigated in real-time without overwhelming the network with overhead.

These challenges motivate the development of machine learning-driven IDS frameworks that are optimized for performance, energy efficiency, and adaptability [22].

1.2 Attacks and countermeasures in wireless sensor networks (WSNs)

Wireless Sensor Networks (WSNs) consist of low-power, resource-limited sensor nodes deployed in potentially hostile environments, making them vulnerable to security threats. These are broadly categorized into passive and active attacks, primarily affecting confidentiality, integrity, availability, and authenticity. In Table 1, we discussed about following attack and their countermeasures;

2 Materials and methods

Prasad et al. [1], along with other researchers, examine the critical security challenges faced by Mobile Ad-Hoc Networks (MANETs) due to their decentralized, infrastructure-less, and highly dynamic nature, which exposes them to a variety of attacks, including black hole attacks, packet dropping, and spoofing. The authors propose an intelligent

intrusion detection system (IDS) specifically designed for MANET environments, incorporating machine learning techniques and efficient feature selection methods, such as Bayesian and rough set approaches, to enhance detection accuracy by focusing on the most relevant features. This strategy minimizes computational overhead, an essential consideration given the limited resources of MANET nodes. Additionally, Prasad et al. introduce a performance reliability evaluation model, assessing the IDS's effectiveness through key indicators like detection accuracy, memory usage, and energy consumption, thus ensuring the system is both robust and efficient. Their model demonstrates substantial improvements over existing IDS frameworks by achieving high detection rates with low false positives, an essential aspect in rapidly changing network topologies. Benchmarking against other IDS models underscores the system's adaptability and efficiency, marking this approach as a promising advancement for secure and reliable intrusion detection in decentralized MANETs.

Prasad et al. [7], alongside other researchers, explore a refined approach for intrusion detection in MANETs that leverages efficient feature selection combined with Bayesian and Rough set techniques. Given the dynamic and resource-constrained nature of MANETs, selecting relevant features is crucial to improve detection speed and reduce computational demands. The authors propose a model that selectively identifies critical features, enabling the intrusion detection system (IDS) to focus on a compact, informative dataset. This approach minimizes memory and processing overhead, which is essential in MANET environments where resources are limited. Prasad et al. demonstrate that, through their optimized feature selection process, the IDS achieves higher detection accuracy with a significant reduction in false positives. Their experiments reveal that this methodology not only enhances the precision of intrusion detection but also ensures that the IDS remains efficient and sustainable in real-time applications, addressing key limitations in traditional IDS frameworks for MANETs.

Wang et al. [2] present a comparative analysis of key MANET routing protocols using the NS3 simulation platform, evaluating AODV, OLSR, and DSDV under various

Table 1 Attacks and countermeasures in WSNs

Attack Type	Impact	Typical Countermeasures
Blackhole	Packet drop, network disruption	Watchdog, trust routing, multipath routing
Wormhole	Route hijack, topology distortion	Packet leashes, directional antennas, time-of-flight checks
Sybil	Node impersonation, protocol disruption	Resource testing, key pre-distribution, authentication
Hello Flood	Misleading neighbor discovery	Bidirectional link checks, signal strength, authenticated HELLOs
Selective Forwarding	Partial data loss	Multipath routing, monitoring, acknowledgments
Sinkhole	Traffic redirection, eavesdropping	Geographical routing, anomaly detection, authenticated metrics
Denial of Service (DoS)	Resource exhaustion	Rate limiting, sleep scheduling, MAC-layer defense

network conditions. By focusing on metrics such as packet delivery ratio, end-to-end delay, and routing overhead, the study explores each protocol's adaptability to MANET-specific challenges like high mobility and frequent topology changes. Findings suggest that AODV, with its reactive approach, performs best in dynamic networks, whereas OLSR is advantageous in more stable settings due to its proactive structure and lower overhead. This analysis highlights the importance of choosing appropriate routing protocols based on network needs and shows NS3 as a valuable tool for assessing protocol performance in MANETs.

Kurniawan et al. [10] perform a comprehensive performance analysis of AODV, OLSR, and DSDV routing protocols in Mobile Ad-Hoc Networks (MANETs) using the NS3 simulation framework. The study focuses on evaluating key performance metrics such as packet delivery ratio, end-to-end delay, throughput, and routing overhead under various network conditions. By simulating different scenarios, the authors assess how each protocol reacts to changes in node density and mobility patterns. The results indicate that AODV excels in scenarios with high node mobility, offering a superior packet delivery ratio due to its on-demand routing capability. In contrast, OLSR shows consistent performance in stable environments, while DSDV exhibits higher overhead and latency. This analysis highlights the critical role of protocol selection based on specific network characteristics, providing insights for optimizing MANET performance in diverse applications.

As'adi et al. [6] introduce a novel statistical method for detecting wormhole attacks in Mobile Ad-Hoc Networks (MANETs), a significant security concern due to their unique characteristics. The researchers emphasize the difficulty of identifying such attacks, which exploit the dynamic nature of MANETs by creating a malicious tunnel between two distant nodes. Their proposed method employs statistical analysis to monitor and analyze packet transmission patterns, allowing for the identification of anomalies that suggest the presence of a wormhole attack. By establishing a baseline of normal network behavior, the method effectively differentiates between legitimate and suspicious activities. The authors demonstrate the efficacy of their approach through simulations, highlighting improvements in detection rates compared to existing techniques. This research contributes to enhancing the security framework of MANETs by providing a reliable means of identifying and mitigating the risks associated with wormhole attacks, thereby bolstering network integrity and reliability.

Vemuri and Mirkar [18] conduct a performance comparison of various routing protocols in Mobile Ad-Hoc Networks (MANETs), focusing on AODV, DSDV, and OLSR. The study evaluates these protocols using metrics such as packet delivery ratio, throughput, end-to-end delay, and routing overhead. By simulating different network scenarios with

varying node densities and mobility patterns, the authors provide insights into the strengths and weaknesses of each protocol. The results indicate that AODV performs best in highly dynamic environments due to its on-demand routing nature, while OLSR offers better performance in stable conditions with lower routing overhead. The findings contribute to the understanding of routing protocol selection in MANETs, emphasizing the need for context-specific evaluations to enhance network performance.

Singla and Jain [20] present a comprehensive comparison of routing protocols for Mobile Ad-Hoc Networks (MANETs) in real-world scenarios using the NS3 simulation environment. The study specifically analyzes the performance of AODV, OLSR, and DSDV by assessing key performance indicators such as packet delivery ratio, average end-to-end delay, and control overhead under varying node mobility and traffic conditions. The authors highlight the practical implications of their findings, demonstrating how the choice of routing protocol significantly impacts network efficiency and reliability. Their results reveal that AODV excels in environments with high mobility, providing a superior packet delivery ratio, whereas OLSR maintains lower overhead in stable conditions. This work underscores the importance of context-sensitive evaluations in protocol selection, paving the way for more effective deployment of MANETs in real-world applications.

Abirami et al. [14] propose an enhanced intrusion detection system (IDS) specifically designed to address routing attacks in Mobile Ad-Hoc Networks (MANETs). The authors emphasize the challenges posed by dynamic network topologies and the necessity for real-time detection mechanisms. Their IDS incorporates advanced techniques for identifying malicious routing behaviors, such as black hole and gray hole attacks. Through extensive simulations, the system demonstrates improved detection rates and reduced false positives, contributing significantly to the overall security of MANETs.

Rajankumar et al. [9] conduct a comparative study and simulation of the AODV routing protocol in Mobile Ad-Hoc Networks (MANETs) using both NS2 and NS3 simulators. The research focuses on evaluating the protocol's performance metrics, such as packet delivery ratio, end-to-end delay, and routing overhead across different network scenarios. The findings indicate that while both simulators provide valuable insights, NS3 offers more accurate modeling capabilities due to its advanced features and flexibility. This study highlights the importance of simulation tools in assessing routing protocols, ultimately guiding improvements in MANET performance.

Fu et al. [15] explore the security vulnerabilities of the AODV protocol in Mobile Ad-Hoc Networks (MANETs), particularly focusing on black hole attacks. The authors identify how these attacks can severely disrupt communication by

falsely advertising the shortest path to a destination, leading to packet loss. To mitigate this threat, they propose enhancements to the AODV protocol that involve implementing a monitoring mechanism to detect suspicious node behavior. Their simulations demonstrate that the enhanced AODV protocol significantly reduces the impact of black hole attacks, resulting in improved packet delivery rates and overall network reliability, thereby strengthening MANET security.

Ahmed and Al-Asadi [23] present a tri-class method to investigate the impact of the number of nodes and sinks on the packet reception ratio in Mobile Ad-Hoc Networks (MANETs) routing protocols. Their study categorizes scenarios based on varying node and sink densities, allowing for a detailed analysis of how these factors influence network performance. By employing simulation techniques, they reveal that an optimal balance between nodes and sinks enhances the packet delivery ratio significantly. The findings underscore the importance of network configuration in optimizing routing protocols, providing insights that can guide future MANET designs and deployments.

Di Mauro et al. [17] conduct a critical review of supervised feature selection techniques in network intrusion detection systems (IDS). The authors highlight the significance of selecting relevant features to improve detection accuracy while minimizing false positives. They examine various algorithms and approaches, discussing their effectiveness in different network environments. The study emphasizes that optimal feature selection not only enhances the performance of IDS but also reduces computational overhead, which is crucial in real-time scenarios. By analyzing existing methodologies, the authors provide valuable insights and recommendations for improving intrusion detection frameworks in both wired and wireless networks.

The paper by Mitrokotsa et al. [24] explores the use of classification algorithms for intrusion detection in MANETs, focusing on the impact of cost sensitivity and model selection on detection performance. The study evaluates several machine learning algorithms, including support vector machines (SVM) and decision trees, to identify malicious activity in dynamic and resource-constrained MANET environments. It highlights the trade-offs between detection accuracy and computational overhead, emphasizing the importance of selecting models that balance these factors. The findings demonstrate that cost-sensitive learning improves the effectiveness of IDS by minimizing false positives while maintaining high detection rates, making it suitable for MANETs where resource optimization is critical. This work provides valuable insights into developing robust and efficient IDS for decentralized and rapidly changing networks.

Zhang et al. [25] explore the application of MANET routing protocols in Vehicular Ad Hoc Networks (VANETs) using the NS3 simulation platform. The study evaluates key routing protocols, including AODV, DSR, and OLSR,

in dynamic and high-mobility VANET environments. Through simulations under various traffic scenarios, the authors analyze metrics such as packet delivery ratio, end-to-end delay, and routing overhead. The findings highlight that while MANET protocols can be adapted for VANETs, their performance is highly influenced by network density and mobility patterns. The paper emphasizes the need for routing solutions specifically designed to address the challenges of VANETs, including high-speed mobility and frequent topology changes, while showcasing the utility of NS3 for VANET research.

Shettar et al. [26] present an intrusion detection system leveraging Multi-Layer Perceptron (MLP) and chaotic neural networks. The study evaluates the models' effectiveness in identifying malicious activities, focusing on their accuracy and robustness. Simulation results demonstrate improved detection rates, showcasing the potential of chaotic neural networks for intrusion detection in dynamic environments.

Elmahdi et al. [19] propose a secure data forwarding mechanism for MANETs using homomorphic encryption to counter blackhole attacks. The approach ensures data confidentiality and reliability while maintaining network performance. The results demonstrate the method's effectiveness in mitigating blackhole attacks and improving secure communication in MANETs.

Sharma et al. [27] analyze wormhole attack formation in MANETs using the Dynamic Source Routing (DSR) protocol, simulated via NS3. The study examines the impact of wormhole attacks on network performance metrics like packet delivery and latency. Results highlight the vulnerability of DSR to such attacks, emphasizing the need for enhanced security measures.

Purohit et al. [28] conduct a simulation-based analysis of Black Hole and Jellyfish attacks on MANETs using NS3. The study evaluates the impact of these attacks on network performance metrics, such as throughput and delay, highlighting their disruptive effects. The findings underscore the importance of developing robust security measures to counter these threats.

El Haoussi et al. [29] propose a method to detect black hole attacks in MANETs using a baiting and timer technique integrated with the AODV protocol. The approach effectively identifies malicious nodes and mitigates attack impacts, enhancing the security and reliability of the network.

Nabou et al. [30] evaluate the performance of MANET routing protocols, AODV and OLSR, under black hole attacks using NS3. The study compares the protocols' effectiveness in terms of packet delivery ratio and network performance, revealing that AODV is more vulnerable to black hole attacks compared to OLSR.

Mobile Ad-Hoc Networks (MANETs) face unique security challenges due to their decentralized architecture,

dynamic topology, and resource-constrained devices. These networks are highly vulnerable to threats such as black hole attacks, spoofing, packet dropping, and wormhole attacks [8]. Addressing these issues requires advanced Intrusion Detection Systems (IDS) tailored to the specific needs of MANETs. Unlike traditional networks, MANET IDS must operate efficiently under limited computational and energy resources while maintaining high detection accuracy [26].

Feature selection is a critical aspect of MANET IDS design. Techniques such as Bayesian models and rough set approaches are employed to identify and prioritize relevant features. This reduces the computational load and allows the IDS to function effectively in resource-constrained environments [17]. By focusing only on the most impactful features, these systems enhance detection accuracy and minimize false positives. Additionally, machine learning methods, including Multi-Layer Perceptron (MLP) models, are integrated to improve threat detection capabilities. These models analyze the patterns of network activity, distinguishing between normal and malicious behaviors with high precision [10].

Routing protocols also play an essential role in MANET security. Protocols like AODV, OLSR, and DSDV are commonly evaluated for their performance in diverse network scenarios. AODV, for example, is well-suited for high-mobility environments due to its reactive nature, which dynamically discovers routes as needed. Conversely, OLSR, a proactive protocol, pre-establishes routes and is more effective in stable networks [20]. Research using NS-3 simulation has demonstrated how these protocols perform under various attack scenarios, highlighting the importance of context-specific protocol selection for secure communication [27].

To mitigate specific threats, such as black hole and wormhole attacks, advanced IDS mechanisms deploy statistical analysis and monitoring strategies. Secure data forwarding techniques, including homomorphic encryption and chaotic neural networks, ensure the integrity and confidentiality of transmitted data [9]. These approaches enhance the overall security framework of MANETs, addressing their inherent vulnerabilities while supporting scalability and adaptability to dynamic conditions. In Table 2, we have summarized the recent IDS techniques in MANET.

3 Technical foundations

3.1 Security strategies

The dynamic and decentralized nature of Mobile Ad-Hoc Networks (MANETs) significantly influences the performance and effectiveness of intrusion detection methods. Achieving high performance in such networks remains a challenging task due to their inherent characteristics, such as mobility and lack of fixed infrastructure. Security

mechanisms in MANETs are generally classified into three broad categories. This study provides a comprehensive analysis of these categories, emphasizing their limitations and challenges in addressing the security needs of MANET environments. These insights aim to highlight the complexities and opportunities for improving intrusion detection in such networks [5].

3.1.1 Communication security frameworks

Communication security in Mobile Ad-Hoc Networks (MANETs) relies heavily on cryptography-based frameworks, which employ sophisticated techniques to safeguard data transmission. These frameworks operate by splitting messages into multiple fragments before sending them, ensuring secure transmission over potentially vulnerable networks. Numerous cryptographic techniques, such as Data Encryption Standard (DES), Triple DES, Advanced Encryption Standard (AES), RSA, and the Diffie-Hellman Key Exchange (DHKE) protocol, are widely utilized for encryption and key management, as highlighted by Usman et al. (2020).

Each node in a MANET independently generates its public and private keys, which are essential for the cryptographic process. Digital signatures play a pivotal role in this framework, as they enable nodes to authenticate data packets efficiently. By verifying the authenticity of the transmitted data, digital signatures enhance trust among nodes, thus maintaining the integrity of communication.

Furthermore, communication security frameworks are often categorized into models such as certificate-based and reputation-based approaches. A certificate-based model leverages a Trusted Third Party (TTP) to establish trust within the same authority domain. The TTP serves as a Certificate Authority (CA), responsible for issuing, verifying, and revoking digital certificates to validate public keys. This Public Key Infrastructure (PKI) ensures the seamless operation of cryptographic security mechanisms.

To strengthen security further, advanced cryptographic techniques like Elliptic Curve Cryptography (ECC), Threshold Cryptosystems (TC), Encrypted Hash Chain (EHC), Message Authentication Code (MAC), and Universal Hash Function (UHF) have been adopted [2]. These advanced methods enhance security by splitting messages into encrypted fragments, making interception and decoding by malicious entities significantly more challenging.

Despite these advancements, certain vulnerabilities persist. Routing attacks such as blackhole and wormhole attacks pose a serious threat to the effectiveness of cryptographic frameworks. Blackhole attacks drop encrypted data packets, disrupting the communication process, while wormhole attacks tunnel encrypted packets through unauthorized

Table 2 Comparative review of recent intrusion detection techniques relevant to mobile Ad-Hoc networks and feature-based detection frameworks

No	Citation	Techniques Used	Research Findings	Limitations
1	Prasad et al. [1]	Intelligent IDS + performance reliability metrics	Enhanced MANET resilience through joint IDS and node trust evaluation	Tailored for MANETs; not directly evaluated in WSNs
2	Prasad et al. [7]	Bayesian classifier + Rough Set + Feature selection	Reduced feature set for lightweight IDS with improved accuracy	Lacks mobility considerations; no deep model comparison
3	Idan et al. [11]	ML classifiers (SVM, RF, DT)	> 95% accuracy; effective for WSN-based datasets	Performance may drop in dynamic networks; lacks deep models
4	Yang et al. [21]	Diffusion model + lightweight classifier	Balanced imbalanced attack data; > 90% accuracy	Real-time deployment challenges in large-scale WSNs
5	Marhoon et al. [31]	Barrier-based ML intrusion detection	Localized monitoring improved boundary intrusion detection	Focused on edge nodes; internal traffic behavior less explored
6	Talukder et al. [12]	SMOTE-Tomek + ML classifiers	High accuracy (~99%); improved learning from minority class	Supervised only; no support for unseen attack types
7	Huang et al. [32]	Hybrid Deep Learning (CNN + RNN) + Federated Learning	Combines privacy with accuracy; resilient to decentralized environments	Communication cost of FL may be high; deep models need more power
8	Behiry & Aly [33]	Hybrid feature reduction (PCA + AI/ML)	Effective dimension reduction; achieved > 96% detection accuracy	Static feature sets; lacks time-sequence modeling
9	Delwar et al. [34]	Survey + Experimental ML Evaluation	Broad comparison of ML for WSN security; practical insights	No new model proposed; general analysis without deployment
10	Mopuru & Pachipala [4]	PCA/SVD + Ensemble ML	Achieved 96% accuracy; reduced energy consumption	Focused on simulated data; no real deployment proof
11	Bukhari et al. [3]	Federated Learning + SCNN-BiLSTM	~99.9% precision/recall; strong privacy and accuracy	Complex; may not suit ultra-low-power WSN nodes
12	Rashid et al. [8]	Federated Learning (FL) for IIoT	Improved scalability and privacy; resilient model sharing	Geared toward IIoT; models might be heavy for WSN
13	Biyyapu et al. [5]	Hybrid sampling + feature aggregation	Enhances NIDS feature learning; reduced overfitting	Focused on general NIDS; WSN-specific tuning not addressed

routes, creating potential breaches in the network. These attacks highlight the limitations of current cryptographic security mechanisms in ensuring robust routing security.

Addressing these challenges necessitates ongoing improvements in cryptographic techniques and the integration of proactive security strategies to mitigate sophisticated attack vectors in dynamic MANET environments. The continuous evolution of cryptographic frameworks is critical for sustaining the security and reliability of MANET communication networks.

3.1.2 Lightweight approaches for intrusion detection

Lightweight techniques (LWT) are efficient and straightforward methods designed to monitor and analyze network or node characteristics with minimal resource usage. These approaches, as described by Subba et al. [35], focus on a limited set of features such as hop count, routing packets, acknowledgments, and threshold values to evaluate node behavior. By employing such simplified metrics, LWTs offer advantages in terms of reduced memory consumption and

computational overhead, making them suitable for resource-constrained environments.

Gaurav and Singh [36] and Hammamouche et al. [37] emphasized that the simplicity of LWTs provides faster processing and adaptability in dynamic networks. However, their minimalistic design has inherent limitations. While LWTs are effective in conserving resources, their ability to detect sophisticated malicious activities, such as routing attacks or node misbehavior, is limited. They often fail to identify malicious nodes that employ complex or stealthy tactics to disrupt network operations.

Some lightweight, learning-based approaches have been developed to address these challenges. These techniques leverage machine learning algorithms to analyze a few critical network features, reducing memory usage while improving detection capabilities. Although these methods alleviate memory constraints, their performance remains insufficient for robustly identifying routing attacks or preventing network vulnerabilities.

The proposed ML-based lightweight approach enhances the intrusion detection process by analyzing a broader set

of 20 network and node-specific features. This approach strikes a balance between computational efficiency and detection accuracy. It achieves higher experimental results while maintaining low resource consumption, addressing the limitations of traditional lightweight techniques. By incorporating more comprehensive data analysis, the proposed method demonstrates significant potential in improving intrusion detection in resource-constrained environments, offering a promising alternative for securing Mobile Ad-Hoc Networks (MANETs).

3.1.3 Heavyweight approaches for intrusion detection

Heavyweight techniques represent a robust and advanced category of intrusion detection approaches that rely on complex rule-based and learning-based methods to identify anomalies within a network [26]. These techniques are designed to handle intricate detection tasks and provide high accuracy in identifying malicious behaviors. However, their complexity comes at the cost of resource consumption and computational overhead.

Unlike lightweight techniques, heavyweight methods require significant memory resources, utilizing both physical memory for data storage and virtual memory for processing. As they analyze large datasets, the computational complexity increases, leading to longer processing times. This makes the classification of transmitted data packets more accurate but also more resource-intensive. The challenge of handling massive datasets can strain the capabilities of Mobile Ad-Hoc Networks (MANETs), which are inherently limited in terms of computational power and memory.

To mitigate these challenges, feature selection techniques [10] are employed to optimize system performance by reducing the dataset size and computational complexity. These methods identify and rank the most relevant features, streamlining the detection process. However, the feature selection process itself can introduce additional computational burdens, as it requires analyzing and prioritizing features, which may not always result in substantial reductions in data volume or system complexity.

Heavyweight techniques are better suited for wired and large-scale networks where resources are abundant and computational demands can be met. Their application in MANETs is limited due to the network's inability to execute extensive computations efficiently. Nonetheless, the proposed method integrates the strengths of heavyweight approaches, achieving higher performance metrics by adopting advanced learning-based techniques. By tailoring these techniques to the constraints of MANET environments, the proposed method demonstrates an effective balance between detection accuracy and resource efficiency, ensuring robust intrusion detection without overburdening the network.

3.2 Characteristics of mobile Ad-Hoc networks

MANET is a dynamic and decentralized network composed of mobile devices such as sensors, smartphones, tablets, and laptops, which act as both hosts and routers. These networks adapt quickly to changing topologies due to node mobility, making them suitable for temporary and rapidly evolving environments like disaster recovery and military operations. However, this mobility introduces challenges, including frequent link failures and routing complexities. Communication in MANET relies on mediums like WiFi, Bluetooth, and infrared, which, despite offering flexibility, are often constrained by interference, limited range, and bandwidth issues.

Resource limitations, such as restricted memory, processing power, and battery life, are defining characteristics of MANET nodes. These constraints significantly impact the design and implementation of network protocols and systems, including intrusion detection mechanisms. Additionally, the absence of centralized infrastructure heightens the network's vulnerability to attacks, requiring robust and efficient security strategies. Understanding these challenges is critical for developing solutions that ensure reliable and secure operations in resource-constrained and highly dynamic MANET environments.

3.2.1 Memory limitations of mobile nodes

This subsection delves into the constrained storage capabilities of mobile nodes in MANET. Mobile nodes, particularly those integrated with microcontrollers, possess limited memory compared to other network devices. Typically, these microcontrollers incorporate several memory chips for data storage and execution. For instance, the CC2538 development kit (cc2538dk) from Texas Instruments (TI) offers a complete system on a chip (SoC) that includes ROM (512 KB) for executable code and RAM (32 KB) for programmed data. Such configurations highlight the modest storage capacities of these nodes, which often rely on external memory for additional space.

Many open-source operating systems, such as Contiki, are optimized for low-power, low-cost microcontrollers. These systems provide basic drivers and facilitate connectivity for IoT and sensor devices. External storage memory chips, with capacities ranging from 512 KB to 32 MB, are frequently used for general-purpose storage or sensor data. Sensor nodes integrate these memory chips for varied functions, depending on their specific roles within the network. While mobile nodes with varying memory capacities are employed in MANET, microcontroller-based sensor nodes typically possess the lowest memory capacity, which imposes additional constraints on their functionality and storage efficiency [18].

3.2.2 Power usage and efficiency

The energy consumption of nodes in a Mobile Ad Hoc Network (MANET) plays a pivotal role in determining the network's overall lifespan. Since nodes are primarily battery-operated, they face significant energy constraints, making power efficiency a critical factor.

To evaluate energy consumption, nodes are analyzed in four states:

- Transmit (t)
- Receive (r)
- Idle (i)
- Sleep (s)

Each state has varying energy requirements, and the energy consumed in each state is computed using the formulas:

$$E_t = P_t \times t_t,$$

$$E_r = P_r \times t_r,$$

$$E_i = P_i \times t_i,$$

$$E_s = P_s \times t_s$$

Where:

- E_t, E_r, E_i, E_s represent the energy consumed in each state.
- t_t, t_r, t_i, t_s denote the time spent in each respective state.
- P_t, P_r, P_i, P_s indicate the power consumed in each state.

The total energy consumption is calculated as:

$$E_{total} = E_t + E_r + E_i + E_s$$

Drain Rate Calculation.

The drain rate (*drain_rate*) is computed using the equation:

$$Lifetime = \frac{(E_{total} \times 3600)}{(V \times T)}$$

where:

- E_{total} is the total energy consumption (in mWh).
- V is the voltage supply.
- T is the time duration (in seconds).

Node Lifetime Estimation.

The estimated lifetime of a node is determined as:

$$Lifetime = \frac{Residualenergy}{Drainrate}$$

where:

- E_{res} is the residual energy of the node.
- *drain_rate* is the computed drain rate.

When the drain rate approaches zero, the node achieves its maximum potential lifetime.

Different types of mobile nodes exhibit diverse energy characteristics. For instance, sensor nodes powered by A Lithium Iron Disulfide (LiFeS₂) batteries with a capacity of 3000 mAh have an extended lifetime suitable for low-energy environments. Research on Power-Aware Heterogeneous AODV (PHAODV) shows improved energy consumption compared to traditional routing protocols like AODV and HAODV. For example, PHAODV achieves an average energy consumption of 0.24 J at an α value of 0.35, where α ranges from 0 to 1, and lower α values indicate reduced energy usage.

The energy capacity of modern batteries varies significantly. Smartphone batteries typically range between 2000 and 4000 mAh, with an average capacity above 3300 mAh. A fully charged 1960 mAh battery can last approximately 14 h, depending on usage. Higher-capacity batteries, with ratings three times the average, can extend operational time significantly, offering a practical solution for energy-intensive MANET nodes. However, rechargeable battery lifespans remain influenced by device energy demands and operating conditions, requiring careful optimization to sustain network performance.

3.2.3 Wireless connectivity and channel distribution

Efficient channel allocation plays a pivotal role in ensuring fair bandwidth distribution among wireless nodes in a MANET. This allocation prevents issues like collisions and contention, enabling smoother communication. Wireless communication technologies such as WiFi operate on unlicensed frequency bands, typically ranging from 2.4 GHz to 5 GHz. The Medium Access Control (MAC) protocol is instrumental in mitigating collision problems within the network by managing access to the shared communication medium.

In MANET, devices like smartphones often serve as mobile nodes, leveraging WiFi for connectivity. WiFi technology offers a cost-effective solution for establishing low-power and low-cost communication networks. It is widely supported across various devices, including laptops, tablets, cameras, and other network-enabled gadgets. Wireless sensors, another integral component of MANET, rely on radio signals to transmit and receive data among nodes within their communication range. These nodes are equipped with capabilities for sensing and data exchange, ensuring seamless interaction between devices in proximity.

A standardized communication protocol is essential for ensuring interoperability among wireless nodes. Technologies like WiFi (IEEE 802.11) and its advanced variants, such as IEEE 802.11n and IEEE 802.11ac, provide this standardization. The newer variants of WiFi technology introduce enhancements in data transmission speed, reliability,

and overall communication efficiency, making them better suited for modern network demands. These advancements enable higher-speed connections and improved network performance, addressing the growing need for efficient wireless connectivity in MANET environments.

3.2.4 Node movement dynamics

The mobility of nodes in MANETs introduces constant changes to network topology, making it highly dynamic and unpredictable. As mobile nodes move, the stability of established connections is impacted, often resulting in frequent disruptions in communication paths. This dynamic topology complicates the process of maintaining accurate route information, as each movement alters the network's structure. The lack of a stable topology poses significant challenges in identifying and maintaining the most efficient path between source and destination nodes.

Node movement dynamics also affect energy consumption, as frequent route rediscovery and path adjustments increase resource usage [38]. Despite extensive research, there is no universally accepted model to accurately predict node mobility or balance energy usage across the network. The absence of such predictive models leads to inefficiencies in route selection and energy management, further complicating the reliability and sustainability of MANET operations.

3.2.5 Analysis of node density

Nodes in a MANET communicate through specific radio frequency ranges, forming connections with neighbors based on their proximity. These connections range from immediate (one-hop) neighbors to multi-hop neighbors, extending to κ hops. The source node typically relies on multi-hop connectivity to reach its destination. However, node density within the network significantly influences the transmission and routing of packets such as Route Request (RREQ), Route Reply (RREP), Route Error (RERR), and Acknowledgment (ACK).

The performance of routing protocols, such as Ad-hoc On-demand Distance Vector (AODV), is particularly sensitive to fluctuations in node density [39]. High or low node density affects protocol efficiency, leading to variations in packet delivery and latency. Furthermore, the reliability of routing is influenced by factors such as node failures, unstable paths, connectivity issues, and node mobility. These dynamics make it critical to analyze node density and its impact on routing protocol performance to enhance the overall efficiency and robustness of MANETs.

3.2.6 Impact of hardware malfunctions

In a MANET, nodes are often deployed in hostile and unpredictable environments, where infrastructure is either

non-existent or has been compromised due to disasters. These conditions make the network highly vulnerable to hardware malfunctions and failures. The failure of a node can be triggered by various factors, including the physical environment, node mobility, battery depletion, and external damage. When a node fails, it can disrupt the communication path, leading to network instability.

Such malfunctions can occur unexpectedly, causing significant communication breakdowns that impair the network's ability to function properly. In these situations, it becomes more challenging to detect malicious activities, as the system's resources are already compromised. Furthermore, the lack of a stable infrastructure increases the complexity of identifying and mitigating threats, as the detection mechanisms depend on the reliability and integrity of the network itself. Thus, ensuring the robustness and reliability of each node and its components is critical for maintaining a secure and functional MANET in such harsh conditions.

4 Proposed method

4.1 Intrusion detection system

The "Intrusion Detection System"(IDS) subsection of the Proposed Method section focuses on advanced strategies designed to enhance network security in Mobile Ad-Hoc Networks (MANETs). Our approach draws from previous research, such as that conducted by Prasad et al., which examined network architectures and attack scenarios, with particular emphasis on Blackhole attacks. A Blackhole attack occurs when a node falsely claims to have the shortest path to a destination, leading to network disruptions and possible data compromise.

Building on this foundational understanding, our method aims to leverage the Binary Relevance Strategy (BRS) within the IDS framework. Unlike other machine learning methods, the Binary Relevance strategy is particularly effective for classification tasks where multiple independent binary classifiers are used. Each classifier addresses a specific aspect of the network's behavior, making it well-suited for managing complex, dynamic environments like MANETs. Previous studies have shown that BRS can perform better than other techniques in handling various network attacks, including Blackhole attacks, by allowing the IDS to make more accurate distinctions between benign and malicious network activities [30].

In our system, the Binary Relevance Strategy employs a set of binary classifiers that independently predict the presence of different types of intrusions based on network data. These classifiers are trained on a dataset containing features extracted from network traffic and node behavior. By focusing on individual attack types, the BRS enables the IDS

to effectively monitor and classify behavior across multiple facets of the network. This approach allows for flexible and scalable detection capabilities that can handle different intrusion patterns in a more granular manner.

Each classifier in the BRS framework is trained using supervised learning, ensuring that it adapts over time based on feedback from observed network behavior. These classifiers update their weights and thresholds continuously, refining their ability to recognize patterns in the network traffic that signal potential security threats. Through this process, the IDS can identify deviations from normal network behavior and flag them as suspicious activities, facilitating early detection of malicious actions.

The architecture of the Binary Relevance Strategy in this research involves several key steps. First, features are extracted from network data, which are then fed into individual binary classifiers. Each classifier independently evaluates whether a given feature set corresponds to a specific type of intrusion. If any classifier detects abnormal behavior, the system generates an alert, allowing for timely intervention.

Overall, the Binary Relevance Strategy employed in our IDS offers a robust solution to MANET security. By using multiple binary classifiers, the system ensures comprehensive coverage of a wide range of potential attacks, while maintaining the flexibility needed to address the unique challenges posed by mobile and dynamic network environments. This method significantly strengthens our ability to detect and mitigate network threats, providing reliable and adaptive security mechanisms for MANETs.

To further enhance the detection capability of the Binary Relevance Strategy (BRS), we integrate a comprehensive dataset that captures various network traffic patterns, node behaviors, and communication features. These features, including packet size, hop count, lifetime, and message type, are key indicators of network health and security. The data used for training the binary classifiers is processed from real-time network traffic, which is critical for training the system to accurately differentiate between benign and malicious activities.

In the context of MANETs, the decentralized nature of the network means that traditional intrusion detection methods, which rely on centralized systems, are less effective. Therefore, each binary classifier in the BRS model operates independently and in a distributed manner, making it highly suitable for mobile, decentralized environments like MANETs. This distributed approach helps minimize the risk of single points of failure, which is a critical consideration in the design of secure and resilient IDS solutions.

Moreover, the Binary Relevance Strategy's ability to handle multiple types of attacks independently is essential for MANETs, where attacks can vary from Blackhole to Sybil, Wormhole, and more. By using distinct classifiers for each attack type, the system can provide more precise detection,

reducing the likelihood of false positives and false negatives. As a result, the IDS can maintain a high level of accuracy, ensuring that only genuine threats trigger alerts.

The system's adaptability is another key feature. As network conditions change and new attack strategies emerge, the binary classifiers are able to update themselves with new data, ensuring the IDS remains effective against evolving threats. This continuous learning process is crucial in dynamic network environments, where the network topology, node mobility, and potential attack strategies constantly shift.

Additionally, the performance of the Binary Relevance Strategy is enhanced by the use of advanced optimization algorithms. These algorithms fine-tune the parameters of each classifier, ensuring that they can quickly adapt to new network conditions and attack scenarios. The combination of supervised learning, feature extraction, and classifier optimization ensures that the IDS maintains high detection rates without compromising system efficiency.

Once the classifiers are trained, the detection mechanism can begin classifying new network traffic in real-time. The output of each classifier is a binary decision indicating the presence or absence of a specific type of intrusion. If multiple classifiers detect abnormal behavior, the system triggers an alarm, alerting network administrators to take appropriate actions. This real-time detection is crucial in preventing the spread of malicious activity across the network, ensuring that the integrity and security of the MANET are preserved.

Furthermore, the integration of the Binary Relevance Strategy with the existing security infrastructure of the MANET provides a robust and scalable security solution. By using multiple classifiers, the IDS is capable of handling a wide range of attack types without being overwhelmed by the complexity of the network or the volume of data. This scalability ensures that as the network grows or changes, the IDS remains effective and efficient.

4.2 Proposed method

4.2.1 Network simulation setup

The research employs NS-3.25 as the network simulator due to its capability to accurately model complex network scenarios, making it suitable for simulating Mobile Ad-Hoc Networks (MANETs) [40]. The simulation environment is meticulously designed to emulate real-world MANET characteristics with high fidelity.

The simulation setup spans a square area of 1000×1000 m, providing ample space to simulate various network scenarios realistically. This area is crucial for evaluating the performance of the Intrusion Detection System (IDS) under different network densities and conditions. Within this space, a variable number of nodes, ranging from

20 to 50, are strategically deployed. This variability in node count enables the simulation of diverse network densities, which is essential for assessing the IDS's ability to handle different network topologies and configurations.

Each node in the simulation operates wirelessly, following the IEEE 802.11 network topology. This topology choice reflects real-world wireless communication protocols used in MANETs, capturing the nuances of communication patterns and data exchange processes. The mobility of the nodes is simulated using the random waypoint mobility model, a widely adopted model in MANET simulations. With nodes moving at a speed of 2 m per second, the simulation accurately reflects the unpredictable and dynamic nature of MANETs, where nodes are continuously changing their positions.

The simulation environment is designed to mimic real-world conditions as closely as possible, allowing for a thorough and accurate evaluation of the IDS's performance. This setup enables the system to be tested in a variety of network scenarios, ensuring that the proposed IDS can effectively detect intrusions and adapt to the challenges presented by diverse MANET conditions. The combination of variable node density, mobility, and network protocol adherence ensures that the simulation provides valuable insights into the robustness and effectiveness of the IDS.

4.2.2 Data acquisition and feature preparation

During the network simulation process, each node generates *.pcap files that capture detailed network traffic information, which is essential for effective intrusion detection and analysis. These files contain various data elements such as packet headers, routing information, communication patterns, and other key network parameters, providing an in-depth view of network behavior. The captured network traffic serves as the raw data for subsequent feature extraction and analysis, forming the backbone of the intrusion detection process.

Before these *.pcap files can be utilized for intrusion detection, they undergo a preprocessing phase where relevant features are extracted. The extracted features are classified into two categories: basic features and derived features. Each feature type provides valuable insights into the network's behavior, helping to identify anomalies or deviations indicative of potential intrusions. A more detailed explanation of these features is as follows:

Basic Features:

- **Duration:** Represents the time difference between the start and end of a packet's transmission. It helps in detecting network delays or congestion that could signal an attack.
- **Protocol:** Indicates the protocol used for communication, such as AODV or modified AODV. Different protocols

may have unique vulnerabilities that can be exploited in attacks.

- **Packet size:** Refers to the length of the transferred packet, represented as an integer. Unusually large or small packet sizes can indicate abnormal traffic or an ongoing attack.
- **Flag:** Indicates the status of the connection, represented by an integer. Abnormal flag values could indicate issues in connection establishment or ongoing attacks.
- **Header length:** Represents the size of the packet header, measured as an integer. Irregularities in header length could suggest packet manipulation, often used in attacks.
- **Hop count:** Refers to the number of nodes between the source and destination, measured as an integer. An unusually high hop count could indicate network manipulation or malicious interference.
- **Lifetime:** Represents the estimated lifetime of a node, recorded as an integer. It helps detect nodes that may have been compromised or are acting abnormally, such as in a Blackhole attack.
- **Message type:** Indicates the message type, such as RREQ, RREP, RERR, or ACK. Recognizing message types helps identify abnormal routing behavior or malicious packet injection.
- **Destination sequence number:** Denotes the possible destination paths that packets may take, represented as an integer. Discrepancies can indicate route poisoning or other malicious attacks.
- **Message sequence number:** Represents the number of nodes the message has passed through. Anomalies in this feature could indicate disruptions in the communication path, often caused by attacks.
- **Stream index:** Identifies if a node is sending the same message to multiple neighbors. Abnormal stream index values could point to flooding attacks or abnormal network behavior.
- **Message transfer mode:** Captures whether the message is broadcast or unicast. A sudden shift in message transfer mode could indicate an attack altering the network communication pattern.
- **Land:** Determines if the sender and originator nodes are the same, different, or unknown. This helps identify spoofing attempts or abnormal communication behaviors.
- **Number of neighbors:** Represents the number of nodes within a node's radio range. A sudden increase or decrease could signal network jamming or node isolation due to malicious activities.
- **Highest flow:** Identifies nodes exhibiting high activity levels, indicating either legitimate or malicious traffic. High flow can signify active nodes participating in attack traffic.
- **Average flow:** Refers to the average amount of data received and transmitted by a node. Significant deviations from the average can indicate unusual network behavior or potential attacks.

- **Lowest flow:** Represents nodes with less data transmission, often indicating a node attempting to remain undetected. Sparse traffic can be a sign of stealthy attacks.
- **Average hop count:** Represents the average number of hops taken by packets. A high average hop count may suggest intentional detours or routing inefficiencies caused by attackers.
- **Number of failed connections:** Indicates the number of Route Error (RERR) messages encountered. A high number of failed connections can signal routing disruptions or intentional network failures.
- **Failed connection rate:** Represents the ratio of failed connections (RERR messages) to total messages. A high rate can indicate attack-induced disruptions in network routes or connectivity.
- **Label:** Categorizes network packets as either normal or malicious. This feature helps classify network behavior and is essential for training intrusion detection models.

This comprehensive preprocessing stage ensures that the extracted features are thoroughly examined and categorized, enhancing the dataset with meaningful and detailed information that captures various aspects of network behavior. These enriched features are crucial for facilitating accurate intrusion detection and analysis in Mobile Ad-Hoc Networks (MANETs).

4.2.3 Intrusion detection system (IDS) implementation

The pre-processed dataset, which contains the meticulously extracted features, serves as the foundational input for a sophisticated Binary Relevance Strategy (BRS) designed explicitly for intrusion detection purposes. This BRS framework is intricately crafted to accommodate the 21 distinct features extracted during the preprocessing stage, providing a rich and comprehensive input space essential for the intrusion detection algorithms to operate effectively.

The Binary Relevance Strategy (BRS) is a multi-label classification technique that transforms the original problem into multiple binary classification tasks. For each feature set, a binary classifier is trained to decide whether a given instance belongs to a particular class, making it highly adaptable to the diverse nature of network behaviors observed in Mobile Ad-Hoc Networks (MANETs). The system processes each feature independently and leverages this to classify each data instance as either normal or malicious. By doing so, the system can capture intricate patterns across multiple feature sets, thereby enhancing the accuracy and sensitivity of the intrusion detection process.

Within this framework, supervised learning techniques are employed to train the system using labeled data, where instances of both malicious and normal network behaviors are provided. This training process plays a crucial role in

optimizing the system's internal weights and biases, allowing it to learn and adapt to complex patterns in the data. It enables the model to discern key distinctions between benign and harmful activities, such as Blackhole attacks and Wormhole attacks, which are significant threats in MANETs. The network continuously adjusts its parameters during training to maximize the classification accuracy while minimizing false positives and false negatives—ensuring the IDS's ability to reliably identify threats without causing unnecessary alarms.

Operating in real-time, the Intrusion Detection System (IDS) continuously monitors network traffic, analysing the incoming data streams for any unusual behaviour or anomalies. When the system detects potential intrusions based on previously learned patterns, it promptly triggers appropriate response mechanisms. These responses can range from sending alert notifications to administrators, to taking more proactive measures such as blocking malicious traffic, filtering suspicious packets, or dynamically adjusting routing paths to isolate affected segments of the network. The goal of these responses is to mitigate potential threats swiftly, limiting the impact of an intrusion while safeguarding the integrity of the network and ensuring uninterrupted communication within the MANET.

To assess the efficacy of the proposed Intrusion Detection System, a thorough performance evaluation is conducted. This evaluation is carried out using a combination of simulated attack scenarios and real-world network traces, providing a comprehensive assessment of the IDS's ability to handle various attack types and network conditions. Performance metrics tailored specifically for intrusion detection systems are used, such as detection rate, false positive rate, and computational efficiency. This exhaustive evaluation ensures that the IDS is not only effective in detecting intrusions but also scalable and capable of operating in a dynamic and resource-constrained environment like a MANET (Fig. 3).

The performance metrics gathered from these evaluations offer critical insights into the system's overall reliability and effectiveness in real-world scenarios. By testing the system against a range of attack strategies, including Blackhole attacks and Wormhole attacks, the research ensures that the Intrusion Detection System can maintain its high detection performance while minimizing errors and delays. This rigorous testing process forms the backbone of the research, offering validation for the IDS's robustness and its practical viability in dynamic Mobile Ad-Hoc Networks.

This comprehensive evaluation process is essential in gauging the overall impact of the Binary Relevance Strategy and its ability to identify malicious behaviours while handling the unique challenges posed by MANET environments. It emphasizes the need for robust, adaptive, and responsive intrusion detection techniques that can operate effectively in uncertain and changing network conditions, ensuring the security and stability of Mobile Ad-Hoc Networks against evolving cyber threats.

Algorithm 1: Binary relevance-based intrusion detection method.

Input:

- num_malicious_nodes: Number of malicious nodes in the network
- num_normal_nodes: Number of normal nodes in the network
- sniffer: Packet analyzer or wireless analyzer
- csv_file: File for storing processed data
- label_encoding: Method for converting symbolic to numeric values
- num_samples: Number of samples in the dataset

Output:

- confusion_matrix: Matrix summarizing the performance of intrusion detection

```

1: Place normal and malicious nodes in a Mobile Ad-Hoc Network (MANET) struc-
   ture.
2: Simulate network activity and capture network traffic (packet capture - .pcap file).
3: Use a packet analyzer or wireless analyzer (sniffer) to extract relevant data.
4: Store the extracted data in a .csv file for further processing.
5: Convert symbolic data into numeric values using the label encoding method.
6: for each sample in the dataset from 1 to num_samples do
7:     Identify the minimum (min_j) and maximum (max_j) values for each feature j.
8:     for each feature i in the sample from 1 to num_samples do
9:         Normalize feature values using min-max normalization:
           
$$x_{i_j} = (x_{i_j} - \min_j) / (\max_j - \min_j)$$

10:    end for
11: end for
12: Remove redundant data in the dataset.
13: Divide the dataset into ten equal sets for 10-fold cross-validation.
14: for each fold (f) in 10-fold cross-validation do
15:     Train a Binary Relevance (BR) classification model on the training sets.
16:     Use the trained BR model to predict the behavior of samples in the test set.
17:     Update the confusion matrix based on predicted and actual behaviors:
18:         if predicted_behavior = testdata_behavior then
19:             Increment the confusion matrix at the true positive (TP) position.
20:         else
21:             Increment the confusion matrix at the false positive (FP) position.
22:         end if
23: end for
24: return confusion_matrix

```

Fig. 3 Training and test set representation in the tenfold cross-validation method



Algorithm 1, an Enhanced Intrusion Detection Method, is pivotal in the proposed research methodology. This algorithm initiates by placing a defined number of normal and malicious nodes within a simulated Mobile Ad-Hoc Network (MANET) environment, replicating real-world network behavior. Once the network structure is established, the simulation generates and captures network activity, storing traffic data in a .pcap file format. This file contains detailed packet-level information, including source and destination addresses, protocols, and data payloads, which are essential for analyzing potential security breaches [27]. A packet analyzer or wireless sniffer is then employed to extract relevant features from the captured data, enabling detailed scrutiny of network behavior. The processed data is stored in a .csv file format for further preprocessing.

To prepare the dataset for machine learning, symbolic data, such as protocol types or flags, is transformed into numeric values using a label encoding method. This step ensures compatibility with machine learning models and standardizes the dataset. Each feature in the dataset is subsequently normalized to bring all values into a consistent range, enhancing model performance. This normalization is achieved by calculating the minimum and maximum values for each feature and applying the normalization formula:

$$X_{ij} = \frac{X_{ij} - \min_j}{\max_j - \min_j}$$

By normalizing the data, the algorithm mitigates potential biases caused by varying scales of feature values. Additionally, redundant data is removed from the dataset to minimize noise and improve the overall quality of the analysis.

To ensure the robustness and generalizability of the intrusion detection model, the algorithm incorporates tenfold cross-validation. The dataset is divided into ten equal subsets, where each subset serves as a test set once, while the remaining nine subsets are used for training. This process

is repeated ten times, allowing the algorithm to evaluate its performance on all parts of the data and reducing the risk of overfitting [26]. During each fold, a Binary Relevance (BR) classification model is trained on the training sets. The BR model handles the intrusion detection task by decomposing it into multiple binary classification problems, each predicting whether a specific behavior (normal or malicious) is present.

Once the BR model is trained, it is tested on the reserved test set. The predicted behaviors are compared with the actual labels, and the results are recorded in a confusion matrix. If the predicted behavior aligns with the actual behavior, the confusion matrix's true positive count is incremented. Conversely, a mismatch leads to an increment in the false positive count. This matrix provides a comprehensive evaluation of the algorithm's performance, capturing metrics such as accuracy, precision, recall, and F1-score.

5 Results and analysis

5.1 Dataset description

In this research, the dataset was meticulously generated from scratch using the NS-3.25 network simulator to emulate a realistic Mobile Ad-Hoc Network (MANET) environment. The simulation was conducted over a $1000 \times 1000 \text{ m}^2$ area with 20 to 50 mobile nodes (this range was chosen after reviewing multiple research papers on intrusion detection systems in MANETs, where 20 to 50 nodes is commonly adopted as a standard node density for realistic and scalable simulations), employing the Dynamic Source Routing (DSR) protocol and the random waypoint mobility model to replicate dynamic network conditions. The nodes were configured to operate over an IEEE 802.11 wireless channel with a transmission range of

250 m and mobility speeds of 2 m/s. The simulation duration varied from 100 to 1000 s to capture diverse network behaviors.

During the simulation, network traffic at each node was recorded using .pcap (packet capture) files, which store detailed packet-level information such as timestamps, source/destination addresses, protocols, and payloads. These .pcap files were parsed using packet analyzers to extract a comprehensive set of 21 features, including Duration, Packet Size, Protocol Type, Hop Count, Message Type, Lifetime, Number of Neighbors, Flow Metrics, and several others. These features were selected based on their relevance to detecting anomalous patterns and were categorized into basic and derived attributes for in-depth behavioral analysis.

The extracted data was pre-processed and normalized using min–max scaling, with symbolic fields encoded numerically to prepare the dataset for machine learning-based classification. Redundant entries were removed to enhance data quality. Each sample is labeled to indicate the nature of the network activity, covering both normal and attack-related behaviors. Simulated attacks such as Wormhole were executed in controlled environments to generate attack data, which was integrated into the final dataset for training and evaluating the proposed Intrusion Detection System (IDS).

This dataset provides a high-fidelity representation of MANET behavior, offering a valuable foundation for training, validating, and benchmarking lightweight and adaptive intrusion detection systems under realistic and resource-constrained network scenarios.

5.2 Data preprocessing

In this research, simulations were conducted utilizing the NS-3 simulator to replicate a Mobile Ad-Hoc Network (MANET) environment with malicious nodes performing blackhole and wormhole attacks, as highlighted in previous studies [1, 7]. NS-3 provides a robust platform for simulating

complex network scenarios, enabling precise modeling of both normal and malicious node behaviors. During the simulations, network activity was recorded at each node in the form of .pcap (packet capture) files, which are commonly used to log detailed network traffic data. These files captured various aspects of network behavior, including packet source and destination, packet size, protocol type, and transmission timestamps.

The data captured in the .pcap files was subsequently processed to extract meaningful features. A packet analyzer tool was employed to parse these files and identify both fundamental features, such as packet size and protocol type, and derived features, such as flow rates and failed connection metrics [11, 21]. This distinction facilitated a comprehensive characterization of network behavior, enabling the detection of anomalies and malicious activities.

Once extracted, the features were systematically organized and stored in a structured database. The database served as a centralized repository, allowing for efficient data retrieval and analysis during the evaluation phase. The entire data preparation process, from the simulation of attacks to

Table 3 Parameters for simulation

Parameter	Values
Network simulator	NS-3.25
Routing protocol	DSR
Simulation area	1000 × 1000 m ²
Addressing mode	IPV4
Number of nodes	20–50
Attack	Wormhole
Channel	Wireless
Mobility of node	Random way point
Speed of node	2 m/s
Network topology	IEEE 802.11
Range of node	250 m
Simulation time	100–1000 s

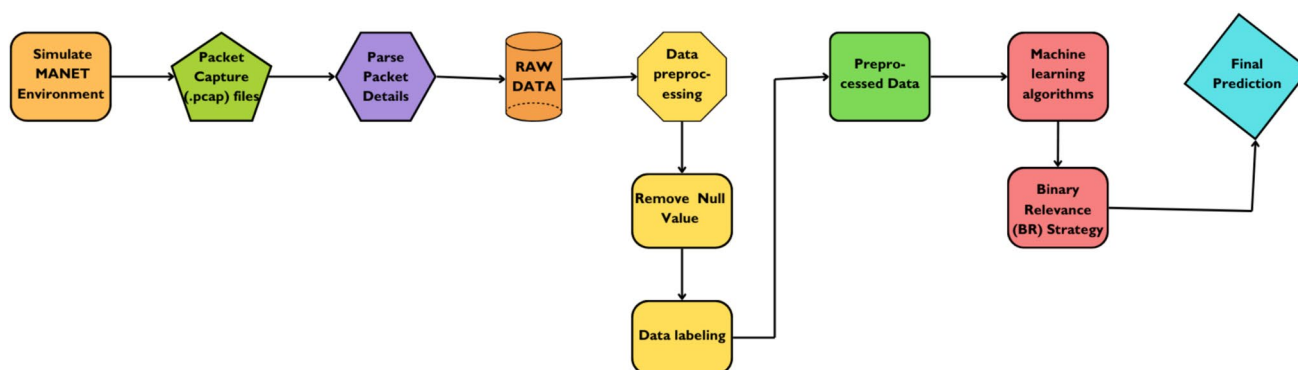


Fig. 4 Proposed approach

the storage of feature-rich datasets in the database, is illustrated in Fig. 4. The initial step involved simulating black-hole and wormhole attacks within the NS-3 environment. The system then generated pcap files for each node, which were analyzed to extract relevant network activity metrics.

The categorization of features into fundamental and derived types provided a strong foundation for subsequent learning and analysis. Fundamental features directly reflect raw network traffic data, while derived features encapsulate higher-level metrics that enhance the ability to characterize complex network behaviors. These features were vital for building robust intrusion detection models capable of differentiating between normal and malicious network activities.

The data preparation methodology underscores the importance of leveraging NS-3 simulations and feature extraction techniques for accurately modeling and analyzing network traffic. By combining simulation-generated data with systematic feature categorization and database organization, this approach lays the groundwork for developing effective intrusion detection systems tailored for dynamic and adversarial network environments.

Table 3, illustrates the simulation parameters utilized in this study, which are specifically tailored for the Dynamic Source Routing (DSR) protocol in a Mobile Ad-Hoc Network (MANET) environment. The simulations, conducted using the NS-3.25 simulator, employed the DSR protocol for normal network nodes [5], while a modified version of

the DSR protocol was integrated to address and mitigate malicious activities such as wormhole attacks [18]. The network setup incorporated realistic conditions, including node mobility modeled using the Random Waypoint mobility model, a node speed of 2 m/s (approximately 7.2 km/h, representing human walking speed), and a simulation area spanning $1000 \times 1000 \text{ m}^2$ [1].

To ensure precise modeling, the simulation included 20–50 nodes, each assigned IPv4 addresses, and a wireless communication channel adhering to the IEEE 802.11 topology [14]. Each node maintained a communication range of 250 m, enabling efficient data exchange and network connectivity. The simulation time varied between 100 to 1000 s, offering flexibility to analyze network behaviors over different durations [15].

Furthermore, an intrusion detection mechanism was implemented at the node level, employing a plug-and-play design to seamlessly integrate into the network without introducing significant routing overhead [17]. The parameters collectively ensured a comprehensive and realistic evaluation of the proposed methods for detecting and mitigating network attacks, enhancing the reliability of the results [23].

The Table 4 provides details of the feature set, outlining the attributes and their corresponding types. These features were identified through an in-depth review of existing mechanisms that utilize network parameters, as listed in Table 4. Key parameters include time delay [19], hop count energy

Table 4 Details of the features

Feature Type	Feature Name	Description	Type
Basic	Duration	Difference in transmission time	Real
	Protocol	Routing protocol type (DSR & modified DSR)	String
	Packet size	Size of the transmitted packet	Integer
	Flag	Connection state indicator	Integer
	Header length	Length of the packet header	Integer
	Hop count	Total nodes traversed between the source and destination	Integer
	Lifetime	Predicted lifespan of a node	Integer
	Message type	Types of messages (RREQ, RREP, RERR, Ack)	String
	Destination sequence number	Possible routes to a destination	Integer
	Message sequence number	Number of intermediate nodes a message passes through	Integer
	Stream index	Repetition of messages sent to neighbouring nodes	Integer
	Message transfer mode	Types of transmission (broadcast/unicast)	Binary
Derived	Land	Relation between sender and originator (same, different, unknown)	Integer
	Number of neighbours	Nodes within communication range	Integer
	Highest flow	Node in a densely active area	Integer
	Average flow	Average signals sent and received by the node	Real
	Lowest flow	Node in a sparsely active area	Integer
	Average hop count	Mean number of hops per transmission	Real
	Number of failed connections	Count of RERR messages	Integer
Label	Failed connection rate	Ratio of RERR messages to total messages	Real
	Classification	Classification of normal and malicious packets	String

or node lifetime message types such as RREQ, RREP, and others, destination sequence numbers, message sequence numbers, and stream indices [26]. Additional parameters, including the number of neighbors, message flow, and data packet information, were also considered [19].

This study includes these parameters and analyzes a total of 20 attributes, including a label for each sample. The feature set is divided into 12 basic features and 8 derived features, with the latter calculated from the basic attributes. When a node enters the network, neighboring nodes extract only its basic features. The proposed method has been trained on a dataset consisting of basic features and has demonstrated superior performance compared to certain existing IDS models.

5.3 Experimental results and performance reliability evaluation

5.3.1 Performance measures

The performance of the work can be assessed using various statistical metrics, which are derived from key parameters in the confusion matrix, such as True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). The confusion matrix reflects the actual positive samples (TP + FN) and negative samples (TN + FP). It also highlights the predicted positive samples (TP + FP) and negative samples (TN + FN). The following equations define the effectiveness of the intrusion detection methods based on these parameters.

Equation 1 defines the True Positive Rate (TRP), also known as Recall, which measures the proportion of actual positive instances that are correctly identified by the model. It reflects the model's ability to detect positive (intrusion) cases and is calculated as:

$$\text{TRP} = \frac{TP}{TP + FN} \quad (1)$$

Equation 2 defines the False Positive Rate (FPR), which quantifies the proportion of normal (negative) instances that are incorrectly classified as positive (attacks) by the detection system. A lower FPR indicates better reliability in distinguishing between benign and malicious traffic:

$$\text{FPR} = \frac{FP}{FP + TN} \quad (2)$$

Equation 3 defines the False Negative Rate (FNR), which measures the proportion of actual positive instances that were incorrectly classified as negative by the model. A high FNR indicates that many attacks are being missed.

$$\text{FNR} = \frac{FN}{TP + FP} \quad (3)$$

Equation 4 defines Precision, which measures the proportion of positive predictions that are actually correct. It reflects the system's ability to avoid false alarms by accurately identifying only true intrusions.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

Equation 5 defines the F-measure (or F1-score), which is the harmonic mean of Precision and Recall. It provides a balanced metric that considers both false positives and false negatives, making it especially useful when the class distribution is uneven or when both false positives and false negatives are costly.

$$\text{F-measure} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Recall} + \text{Precision}} \quad (5)$$

Equation 6 defines Accuracy, which measures the overall correctness of the intrusion detection system by computing the ratio of correctly classified instances (both true positives and true negatives) to the total number of samples. It gives a general indication of how well the model performs across all classes.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

Equation 7 defines the Packet Delivery Fraction (PDF), which represents the ratio of the total number of packets successfully received by the destination nodes to the total number of packets transmitted by the source nodes. It reflects the efficiency and reliability of the network routing protocol under evaluation.

$$\text{PDF} = \frac{\text{Number_of_received_packets}}{\text{Number_of_sent_packets}} \quad (7)$$

Equation 8 defines the Packet Loss Ratio (PLR), which measures the proportion of packets lost during transmission relative to the total number of packets sent. A lower PLR indicates higher transmission reliability and network stability.

$$\text{PLR} = \frac{\text{Number_of_lost_packets}}{\text{Number_of_sent_packets}} \quad (8)$$

Equation 9 defines the Packet Delivery Ratio Rate (PDR), which quantifies the ratio of packets successfully received at the destination to the total number of packets generated at the source. It is a key indicator of network throughput and delivery efficiency.

$$\text{PDR} = \frac{\text{Received_packets_by_the_destination}}{\text{Generated_packets_by_the_source}} \quad (9)$$

Intrusion isolation serves as a performance evaluation parameter, calculated based on the Confidence on Attack

(CoA) ratio. The Defense Rate, equivalent to detection accuracy, measures how effectively the system detects intrusions. Overhead is assessed by comparing the total number of routing packets generated to the total number of packets transmitted (both generated and forwarded) in the network. Normalized Routing Load (NRL) determines how many routing packets are transmitted for each data packet successfully delivered to the destination, including hop-wise transmission of routing packets. Throughput represents the rate at which data is successfully received at the destination, measured in bits per second. The Packet Drop Rate (PDR) is identical to the Packet Loss Rate (PLR), which tracks lost packets. The Packet Delivery Rate (PDRr) is the ratio of received packets to sent packets at the destination, while the Packet Delivery Fraction (PDF) also indicates the ratio of received packets to those sent.

The True Positive Rate (TPR), also known as the Detection Rate (DR) or Recall, measures the system's positive performance, derived from the true positives (TP) and false negatives (FN). On the other hand, the False Positive Rate (FPR) or False Alarm Rate (FAR) reflects the system's negative performance, alongside the False Negative Rate (FNR), which represents the underperformance in detecting intrusions. The following tables highlight the best-performing intrusion detection methods. The performance evaluation model calculates reliability by multiplying positive performance values as per Eq. 1, and reversing it for negative performance (by subtracting from 1). While different works have measured various statistical parameters to assess system performance, a comparative analysis is lacking. These parameters focus on false predictions and packet drop rates, which are essential for determining the network's overall security. Thus, this study assigns equal weight to all the statistical performance metrics.

5.4 Complexity analysis

5.4.1 Performance reliability evaluation

Data preprocessing plays a crucial role in preparing network traffic data for training the Intrusion Detection System (IDS). This stage involves the following tasks: feature extraction, normalization, and redundancy reduction. Each of these steps comes with specific computational complexities:

Feature Extraction: In the feature extraction step, 21 features are extracted from the network traffic data. Let P represent the number of packets in the dataset. The complexity for feature extraction is $O(P)$, since each packet is processed individually.

Normalization: Normalization ensures that each feature contributes equally to the model. During this step, the minimum and maximum values of each feature are calculated

across N samples. The complexity of this operation is $O(N)$, where N is the number of samples.

Redundancy Reduction: To reduce overfitting and improve model performance, redundancy reduction identifies and removes redundant data points. The complexity of this process is $O(N \cdot f)$ where f is the number of features being analyzed.

5.4.2 Simulation setup complexity

Simulating a Mobile Ad-Hoc Network (MANET) using NS-3.25 involves several key stages, including node initialization, simulation execution, and implementing mobility models and communication protocols. The complexity of each stage is as follows:

- **Node Initialization:** The complexity of initializing each network node is $O(n)$, where n represents the number of nodes in the network.
- **Simulation Execution:** The simulation's execution time depends on the number of events generated. NS-3 uses an event-driven scheduler, which handles events with a complexity of $O(E \cdot \log n)$, where E is the number of events and n is the number of nodes.
- **Mobility Models and Communication Protocols:** Implementing mobility models and configuring communication protocols adds complexity, but these tasks are typically linear in relation to the number of nodes and the simulation duration.

5.4.3 Overall computational complexity

When combining the complexities from multiple stages, including MLP training, data preprocessing, and network simulation, the overall computational complexity of the proposed IDS approach can be approximated as:

$$O(E \cdot N \cdot (mh + ho)) + O(P) + O(N + f) + O(E \cdot \log n)$$

The above Equation defines the overall computational complexity of the proposed intrusion detection framework, combining various components that impact performance in Mobile Ad-Hoc Networks (MANETs). The terms represent:

E : number of edges or links in the network graph.

N : number of nodes.

mh : cost of message handling.

ho : overhead of heuristic operations.

P : number of packets processed.

f : number of features considered during detection.

n : input size for sorting or heap operation.

This complexity analysis demonstrates the computational demands of the proposed IDS approach. The study emphasizes the importance of implementing optimization

techniques to handle large-scale environments and extensive traffic data typical in MANETs. Additionally, the preprocessing and training steps have been carefully designed to ensure the robustness and accuracy of the IDS while managing the inherent computational complexities.

5.4.4 Performance evaluation

To assess the performance of the proposed IDS, extensive simulations and experiments were conducted, as referenced by [29]. The evaluation metrics include detection accuracy, false positive rate, and computational efficiency:

- **Detection Accuracy:** The system achieved a high detection accuracy, effectively identifying various types of intrusions in the dynamic MANET environment. The ability of the Multi-Layer Perceptron (MLP) to learn complex patterns from network traffic data contributed significantly to this high level of accuracy.
- **False Positive Rate:** Minimizing false positives is essential for the practical deployment of an IDS. This system demonstrated a low false positive rate, ensuring that legitimate network activities were rarely misclassified as intrusions.
- **Computational Efficiency:** Despite the computational complexity of MLP training and data preprocessing, the proposed IDS was computationally efficient. This makes it suitable for real-time intrusion detection in dynamic and large-scale MANETs. Optimization techniques such as efficient feature extraction and redundancy reduction played a vital role in improving the system's performance.

By employing these optimization strategies, the IDS system successfully addresses both the accuracy and efficiency challenges associated with intrusion detection in complex and dynamic environments such as MANETs.

5.4.5 Time complexity of feature extraction

Feature extraction is a critical step in preparing the network traffic data for training the Intrusion Detection System (IDS). The process involves extracting 21 distinct features from the network traffic data. If we represent the number of packets as P , the time complexity of feature extraction is:

$$O(P)$$

This reflects the fact that each packet is processed individually to extract the relevant features.

5.4.6 Time complexity of the machine learning model (MLP)

Training the Multi-Layer Perceptron (MLP) model involves processing the training data to learn patterns and correlations

for intrusion detection. The time complexity of training an MLP is influenced by the number of training samples N , the number of features f , and the number of iterations T required to converge during training. The time complexity is expressed as:

$$O(N \cdot f \cdot T)$$

Where:

- N is the number of samples in the training set,
- f is the number of features (21 in our case),
- T is the number of iterations or epochs required for training.

This complexity assumes that each iteration processes all samples and all features.

5.4.7 Time complexity of the simulation environment (NS-3)

Simulating a Mobile Ad-Hoc Network (MANET) using NS-3.25 involves several steps, such as node initialization, simulation execution, and mobility model implementation. The time complexity can be broken down into the following components:

- **Node Initialization:** Initializing each node in the network has a complexity of $O(n)$, where n is the number of nodes in the simulation.
- **Simulation Execution:** The time complexity for executing the simulation depends on the number of events E generated and the number of nodes n in the simulation. Since NS-3 uses an event-driven scheduler, the time complexity of managing these events is:

$$O(E \cdot \log_n)$$

where E is the number of events and n is the number of nodes.

Mobility Models and Communication Protocols: The complexity of implementing mobility models and configuring communication protocols is typically linear in relation to the number of nodes and the duration of the simulation. Therefore, the time complexity for these tasks can be considered as:

$$O(n)$$

5.4.8 Evaluation complexity

The evaluation of the IDS performance involves calculating various metrics such as detection accuracy, false positive rate, and computational efficiency [23]. The evaluation

complexity is influenced by the number of test samples N and the number of metrics computed. In general, the time complexity of the evaluation phase can be expressed as:

$$O(N \cdot M)$$

where:

- N is the number of test samples.
- M is the number of evaluation metrics being calculated (e.g., detection accuracy, false positive rate, etc.).

This complexity assumes that each test sample is evaluated against each performance metric.

Overall Complexity.

Combining all the complexities from feature extraction, machine learning model training, and simulation setup, the overall computational complexity for the proposed IDS approach can be approximated as:

$$O(P) + O(N \cdot f \cdot T) + O(E \cdot \log_n) + O(n) + O(N \cdot M)$$

The above Equation represents the overall time complexity of the proposed adaptive intrusion detection framework for Mobile Ad-Hoc Networks (MANETs), accounting for multiple interacting operations involved in packet processing, feature extraction, network management, and classification. Each term represents a specific computational component:

Where:

- P is the number of packets.
- N is the number of samples.
- f is the number of features.
- T is the number of training iterations.
- E is the number of events.
- n is the number of nodes.
- M is the number of evaluation metrics.

This comprehensive complexity analysis highlights the computational demands of the IDS approach, emphasizing the need for efficient implementation and optimization strategies to handle large-scale networks and extensive traffic data in dynamic MANET environments.

5.5 Performance analysis

The evaluated Table 6 presents the reliability and performance of five security mechanisms—SLP (Supervised Learning Protocol), MLP (Multi-Layer Perceptron), BR Strategy (Blackhole Resilient Strategy), K-mean (Clustering-based Approach), and KNN (K-Nearest Neighbors)—against the Blackhole attack in Mobile Ad-Hoc Networks (MANETs). These techniques are assessed on two datasets,

Dataset 1 and Dataset 2, using critical performance metrics such as Precision, Recall, F1-Score, Accuracy, RMSE (Root Mean Square Error), and MAE (Mean Absolute Error). The evaluation is conducted under two statistical categories: Macro Average, which provides an unweighted average across all classes, and Weighted Average, which adjusts the averages based on class distribution.

SLP demonstrates consistent performance with high accuracy across both datasets. On Dataset 1, it achieves a Macro Average F1-Score of 87.54% and Weighted Average Precision of 90.65%, resulting in an overall accuracy of 87% with an RMSE of 18.60 and MAE of 18.01. Similarly, on Dataset 2, SLP performs slightly better, attaining a Macro Average F1-Score of 90.75%, with an overall accuracy of 90.10%. These results indicate the robustness of SLP in detecting Blackhole attacks, as evidenced by its low error rates.

In contrast, MLP shows variability in performance. On Dataset 1, it achieves a Macro Average F1-Score of 85% and an accuracy of 79%, but its RMSE (45.84) and MAE (21.11) are significantly higher compared to SLP, indicating greater prediction errors. On Dataset 2, MLP's performance declines further, with a Macro Average F1-Score of 43% and an accuracy of 86%. The Weighted Average scores also reflect inconsistencies, suggesting that MLP struggles to handle the complexities of Blackhole detection effectively.

The BR Strategy emerges as the most reliable technique in the analysis. It achieves remarkable performance metrics across both datasets, with a Macro Average F1-Score of 98.84% on Dataset 1 and 98.75% on Dataset 2. Its overall accuracy reaches 98% on Dataset 1 and 96% on Dataset 2, coupled with low RMSE (7.60 and 10.00, respectively) and MAE (7.01 and 13.00). These results highlight the BR Strategy's effectiveness in maintaining high detection accuracy while minimizing prediction errors.

The clustering-based K-mean approach exhibits moderate performance. On Dataset 1, it achieves a Macro Average F1-Score of 20% and an accuracy of 61%, but its RMSE (22.00) and MAE (27.00) suggest substantial prediction errors. On Dataset 2, K-mean improves slightly, with a Macro Average F1-Score of 61% and an accuracy of 68%, although error rates remain relatively high. This indicates that K-mean may not be the most suitable technique for complex Blackhole detection tasks.

KNN, the final technique evaluated, shows mixed results. On Dataset 1, its Macro Average F1-Score is 25%, with an accuracy of 60%, and error rates of RMSE 38.00 and MAE 40.00. On Dataset 2, it fares slightly better, achieving a Macro Average F1-Score of 61% and an accuracy of 67%, but still lags behind other techniques like BR Strategy and SLP in terms of reliability and precision. The high RMSE and MAE values across both datasets indicate challenges in accurately classifying attacks.

Table 6 Assessed additional performance of IDS on the generated dataset

Technique	Attack	Type	Dataset	Precision	Recall	F1-Score	Accuracy	RMSE	MAE
SLP	Wormhole	Macro Average	Dataset 1	87.45	86.52	87.54	87.00	18.60	18.01
		Weighted Average	Dataset 1	90.65	90.41	86.02			
	Wormhole	Macro Average	Dataset 2	90.00	89.74	90.75	90.10	19.00	22.00
		Weighted Average	Dataset 2	87.00	85.74	89.75			
MLP	Wormhole	Macro Average	Dataset 1	78.00	81.00	85.00	79.00	45.84	21.11
		Weighted Average	Dataset 1	82.00	89.00	75.00			
	Wormhole	Macro Average	Dataset 2	39.00	49.00	43.00	86.00	54.55	30.05
		Weighted Average	Dataset 2	63.00	72.00	70.00			
BR Strategy	Wormhole	Macro Average	Dataset 1	98.85	97.62	98.84	98.00	7.60	7.01
		Weighted Average	Dataset 1	97.15	97.21	92.02			
	Wormhole	Macro Average	Dataset 2	97.90	96.54	98.75	96.00	10.00	13.00
		Weighted Average	Dataset 2	96.80	95.94	94.85			
K-mean	Wormhole	Macro Average	Dataset 1	30.00	62.00	20.00	61.00	22.00	27.00
		Weighted Average	Dataset 1	64.00	31.00	34.00			
	Wormhole	Macro Average	Dataset 2	39.00	75.00	61.00	68.00	46.00	42.00
		Weighted Average	Dataset 2	86.00	61.00	70.00			
KNN	Wormhole	Macro Average	Dataset 1	39.00	75.00	25.00	60.00	38.00	40.00
		Weighted Average	Dataset 1	91.00	61.00	70.00			
	Wormhole	Macro Average	Dataset 2	35.00	84.00	61.00	67.00	34.00	37.00
		Weighted Average	Dataset 2	92.00	60.00	69.00			

Overall, the table reveals that BR Strategy outperforms the other techniques in all metrics, demonstrating exceptional reliability and accuracy in detecting Blackhole attacks. SLP also performs well, particularly on Dataset 2, making it a strong alternative. MLP, K-mean, and KNN show varying levels of effectiveness but struggle with higher prediction errors and lower accuracy, especially when faced with complex datasets. These results emphasize the importance of selecting robust and reliable security mechanisms for intrusion detection in MANETs, where performance trade-offs can significantly impact network security.

6 Conclusion

This research introduces a comprehensive and adaptive Intrusion Detection System (IDS) specifically designed for the complex, decentralized environment of Mobile Ad-Hoc Networks (MANETs). The proposed framework addresses critical limitations in existing IDS solutions by integrating machine learning, feature optimization, and fuzzy logic to deliver a solution that is both accurate and resource-efficient. Leveraging the Binary Relevance Strategy (BRS), the system effectively decomposes multi-label classification into independent binary classifiers, enabling precise detection of diverse intrusion types such as blackhole and wormhole attacks.

A key innovation of the framework lies in its ability to dynamically adapt to the volatile topologies of MANETs

through the inclusion of a performance reliability evaluation model based on Fuzzy Logic Systems (FLS). This addition allows the IDS to intelligently assess its effectiveness under varying energy, memory, and mobility constraints, ensuring sustained performance in real-world deployment scenarios. The integration of fuzzy logic improves decision-making sensitivity, enabling the system to react effectively to fluctuating network behavior without significantly increasing overhead.

To validate the robustness of the proposed IDS, an exhaustive set of simulations was conducted using NS-3.25, reflecting realistic node mobility, routing protocols (DSR), and diverse network attack scenarios. The network simulation generated granular traffic data captured in .pcap files, which were preprocessed to extract 21 high-impact features. These features were rigorously analyzed and fed into supervised learning models, offering a structured and informed approach to intrusion classification. Through tenfold cross-validation, the IDS demonstrated high detection accuracy, low false positive rates, and consistent performance across dynamic test environments.

Furthermore, the system emphasizes scalability and modularity, ensuring it remains effective even as the network grows or threat types evolve. The ensemble of binary classifiers within the BRS can be updated independently, allowing the IDS to incorporate new threat signatures without retraining the entire system—an important capability in MANETs where threats are often unpredictable and rapidly evolving.

This work also highlights the broader implications of intelligent feature selection in resource-constrained environments.

By focusing on the most informative attributes of network traffic, the system reduces both memory usage and processing time, making it suitable for real-time deployment in lightweight MANET nodes. The findings underscore the importance of using hybrid approaches—merging statistical, rule-based, and machine learning methods—to address the unique security challenges posed by mobile, infrastructure-less networks.

In conclusion, the proposed IDS framework represents a significant advancement in secure MANET communication. It merges theoretical rigor with practical applicability, offering a detection system that is intelligent, adaptive, and resource-aware. Future work could explore integrating real-time threat intelligence sharing between nodes, expanding attack detection beyond known patterns using semi-supervised or unsupervised learning, and enhancing resilience against coordinated, multi-vector attacks.

6.1 Finding

A notable enhancement in this work is the integration of a Fuzzy Logic System (FLS), which allows the IDS to dynamically evaluate its performance under varying network conditions and resource constraints. This ensures a balance between computational efficiency and energy consumption without sacrificing detection accuracy. Key performance metrics such as packet delivery ratio, detection accuracy, and energy efficiency are thoroughly analyzed, demonstrating that the proposed system significantly outperforms traditional IDS frameworks in both reliability and adaptability.

6.2 Strength

The system's strengths include its precise detection capabilities enabled by BRS and an extensive feature set, real-time adaptability with minimal computational overhead, and robust performance even in energy-constrained environments. These advantages make it particularly suitable for mission-critical applications like military operations and disaster response, where reliability and low-latency decision-making are essential.

6.3 Limitation

However, the study also acknowledges some limitations. The current implementation relies on simulation-based validation, which may not capture all the complexities of real-world network behavior. Additionally, the system has only been evaluated on two primary components and depends on manually preprocessed data, which can limit its scalability and generalization.

6.4 Future work

While the proposed IDS demonstrates strong performance in simulated MANET environments, future enhancements

will focus on integrating additional network parameters into the detection model. Variables such as node density changes, route instability metrics, and signal strength variations could provide richer context and improve intrusion classification accuracy. Incorporating such dynamic metrics will enhance the IDS's sensitivity to complex and evolving attack behaviors, particularly in high-mobility scenarios.

A critical advancement will involve automating the data preprocessing pipeline, which currently includes manual steps like packet capture processing, feature extraction, and normalization. Developing an automated pipeline will streamline the deployment of the IDS in real-time environments and enable continuous data ingestion and model updates. This is especially important for MANETs, where manual data handling is impractical due to limited resources and constantly changing topology.

Another area of focus is the exploration of hybrid IDS architectures that combine the strengths of anomaly-based and signature-based approaches. These architectures would allow the system to detect both known and previously unseen attacks, offering improved resilience against zero-day exploits. Incorporating statistical models or rule-based logic alongside supervised learning algorithms could further improve classification robustness and reduce false positives in complex environments. Additionally, future evaluations will involve real-world testing on physical MANET deployments, such as mobile sensor nodes or vehicular networks, to assess performance under real communication noise, variable energy availability, and unpredictable node movement.

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Declarations

Ethics, Consent to Participate, and Consent to Publish Declarations Not applicable.

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